

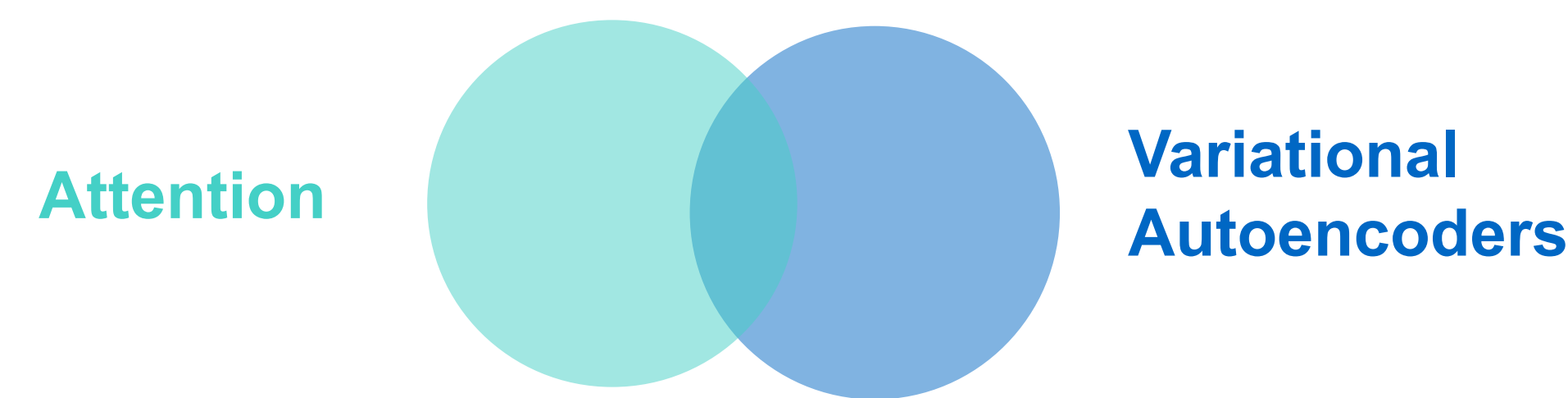
Multi-Head Attention In Single Cell Variational Autoencoders

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Abstract

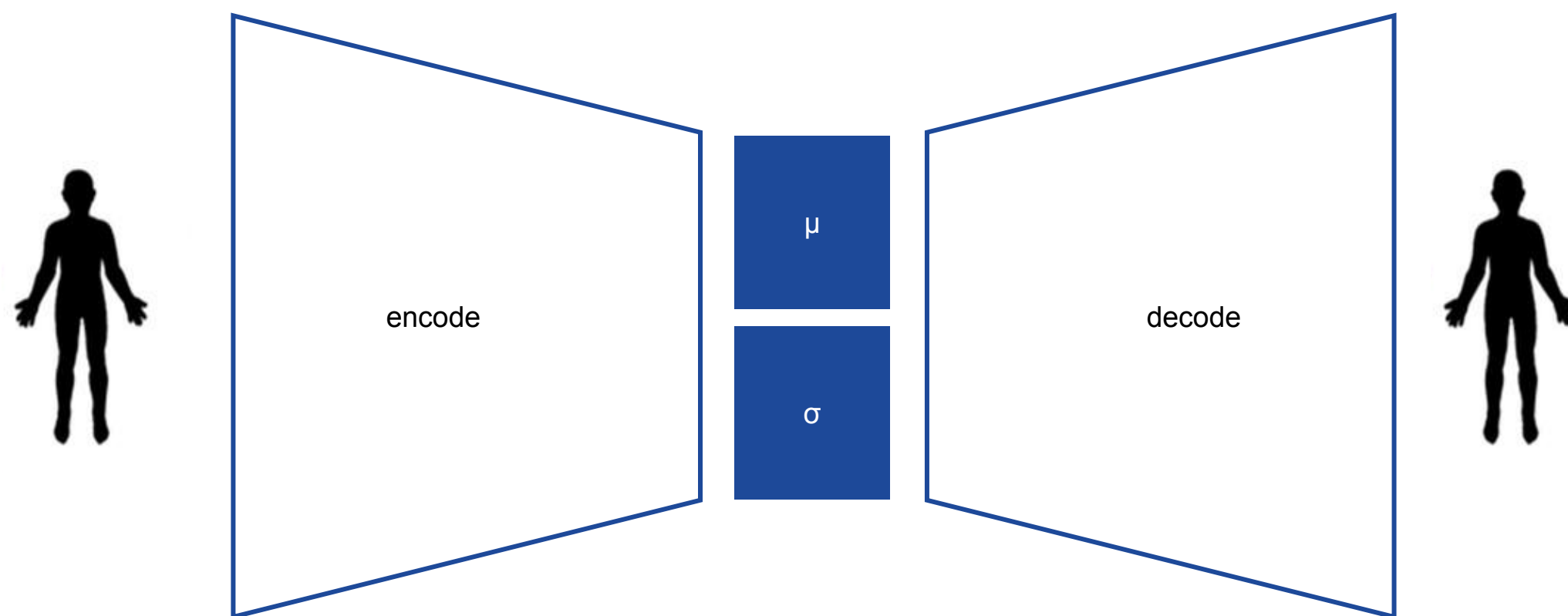
Artificial Intelligence has experienced a massive boom in recent years with the advent of Large Language Models. These models rely heavily on Transformer based architectures, like the Generative Pretrained Transformer or GPT. In the area of single cell transcriptomics however, Transformer based architectures have yet to show the revolutionary success present in language modeling. Variational Autoencoders, another variety of AI architecture, have had some success in the field. This investigation aims to observe the performance impacts of adding the core mechanism of the Transformer, multi-head self attention, into a counts-based transcriptomics Variational Autoencoder.

Project Goals



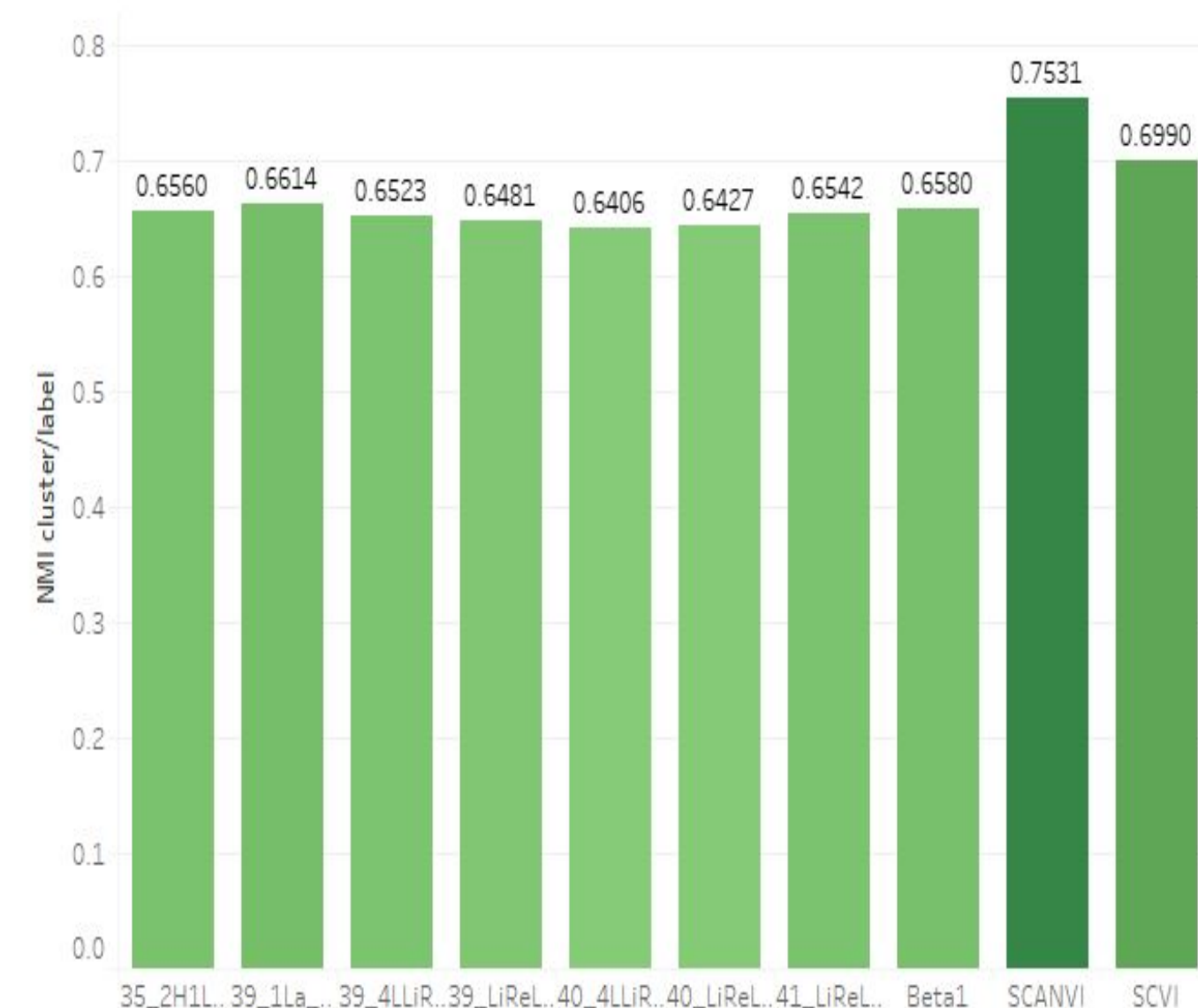
- Test adding multi-head self attention to a single cell human Variational Autoencoder.
- Compare results to linear depth for validation testing.

Variational Autoencoders



- Maps inputs to a latent space usually modeled by a normal distribution.
- Encodes the data, uncovering underlying relationships between data visible in clustering UMAPs.
- Reconstruction(MSE) and KL-Divergence monitor performance of the model.

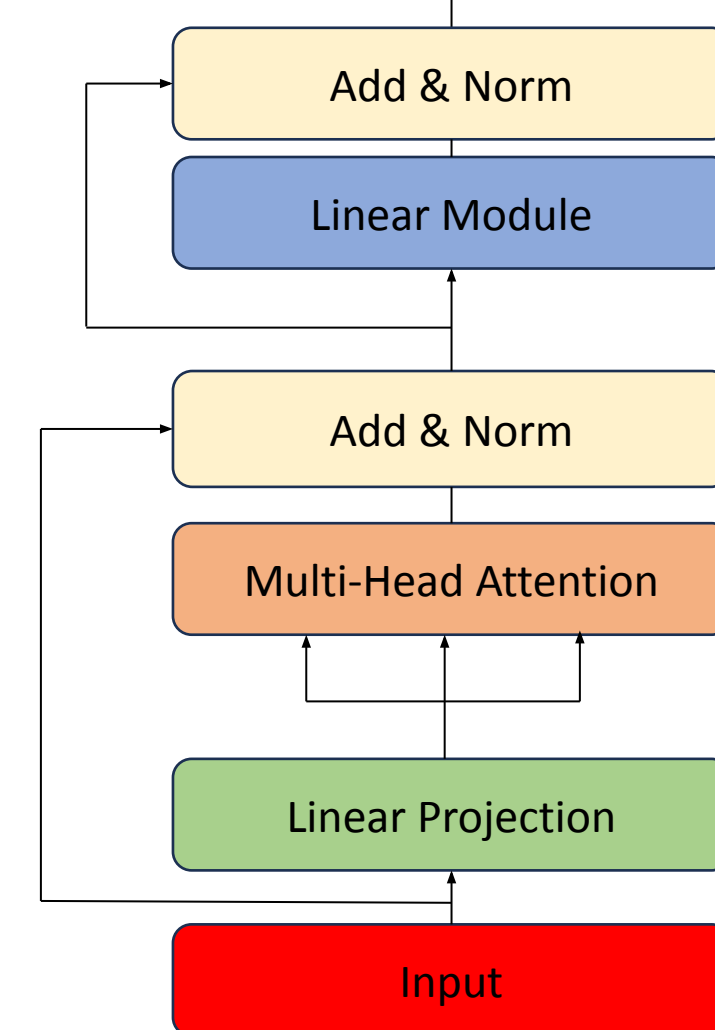
Performance Metrics



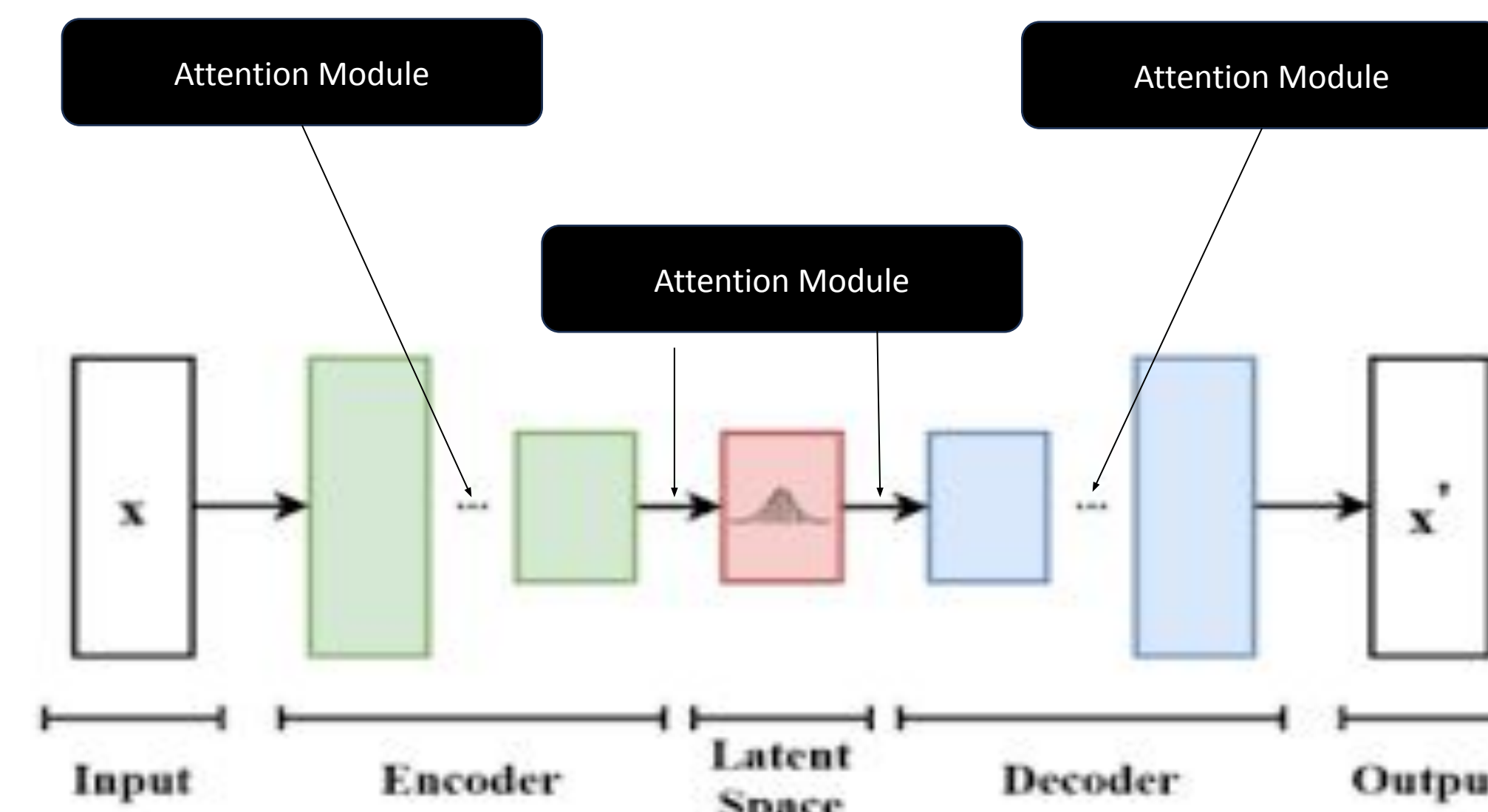
Not significant improvement with Multi-head Attention compared to depth.

Attention Module and Design

Attention Module:



Experiment Design:

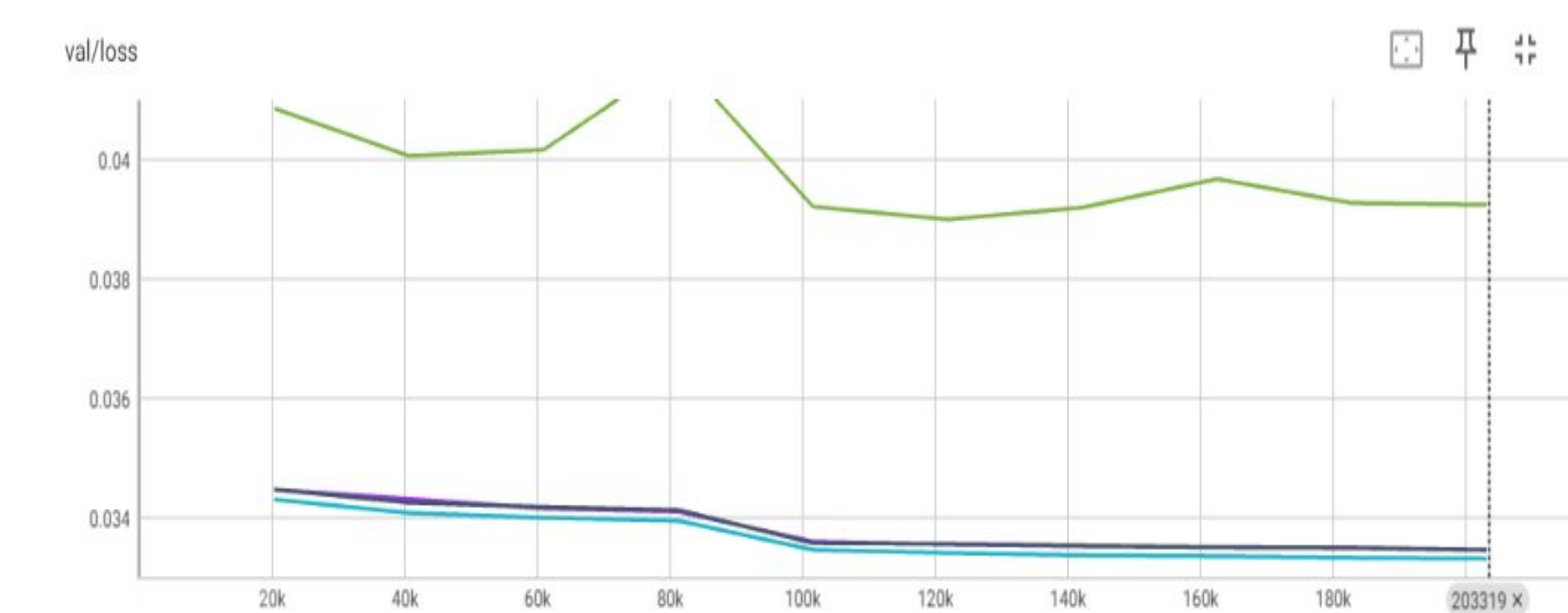


Tested all possible locations and combinations for best results.

- Structure inspired by Transformer Encoder in "Attention is All You Need.", Vaswani, Ashish, et al.
- Optimal structure validated by testing.

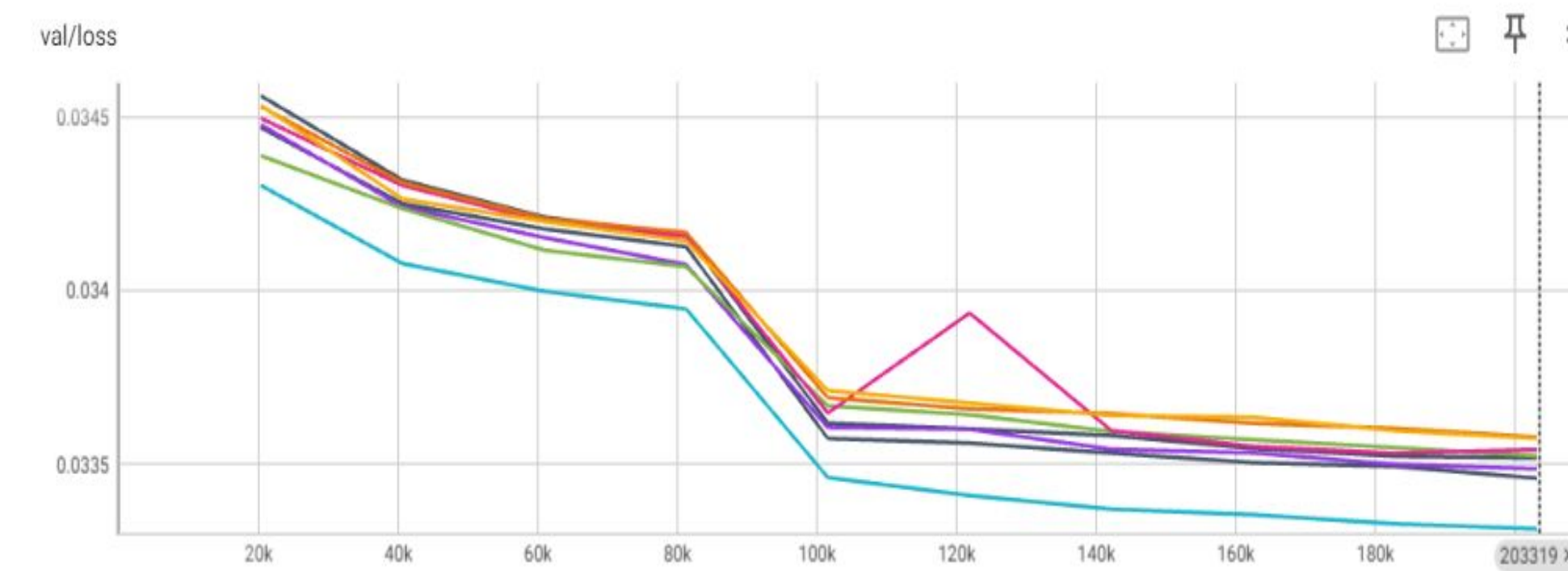
Testing and Results

Step 1: Test Attention Methods



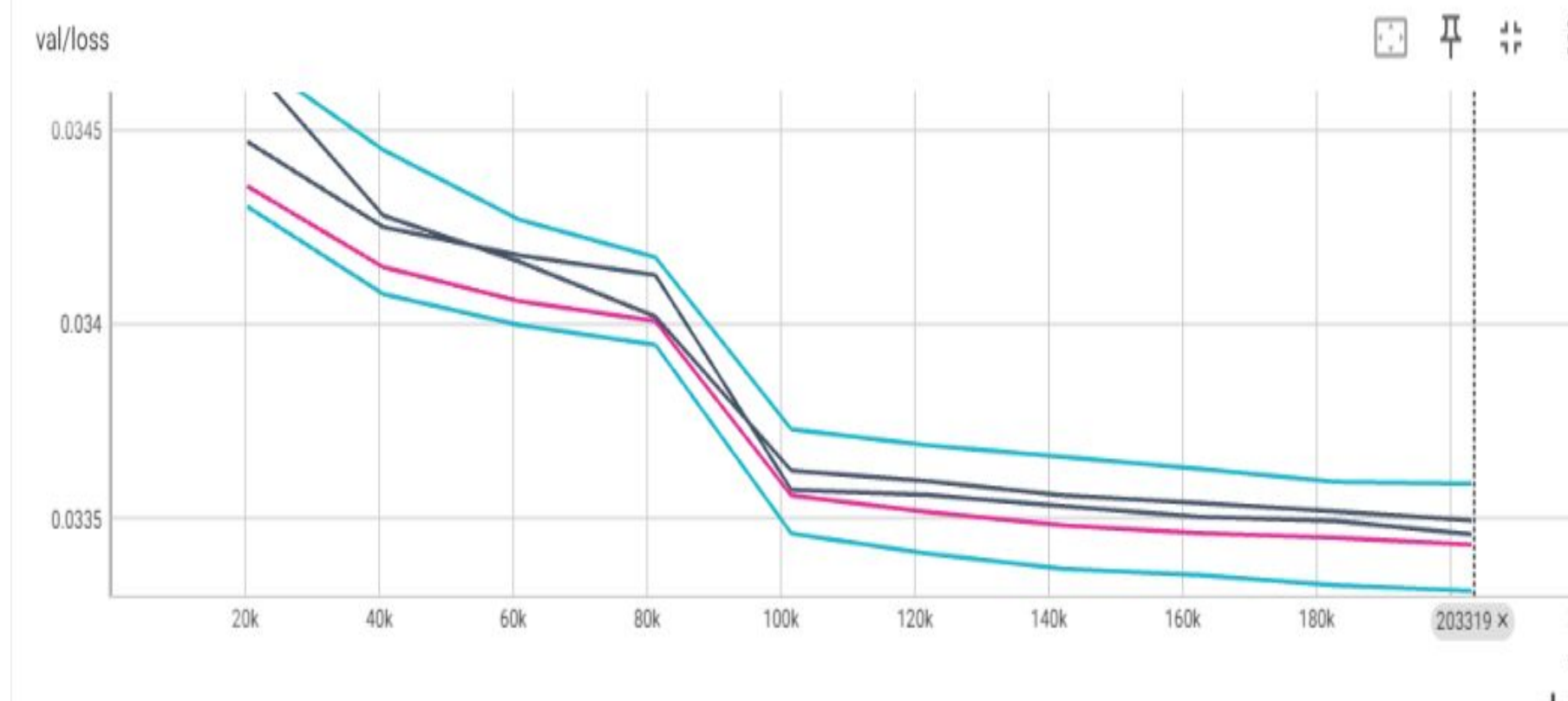
Linear Attention performed best, using 2 attention heads as a baseline.

Step 2: Test Different Number of Heads



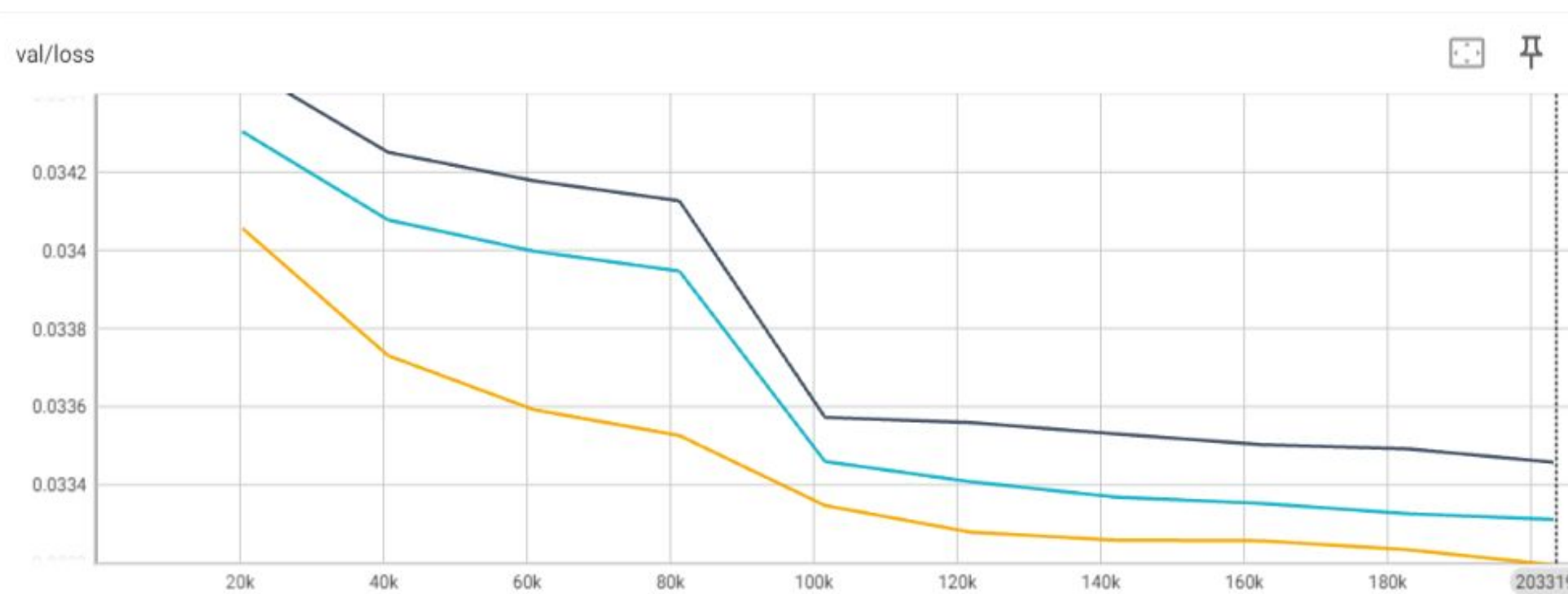
2 attention heads outperformed 4, 8, 16, 32, 64, and 128 heads in the dim=256 encoder layer.

Step 3: Test Linear Depth and Chained Modules



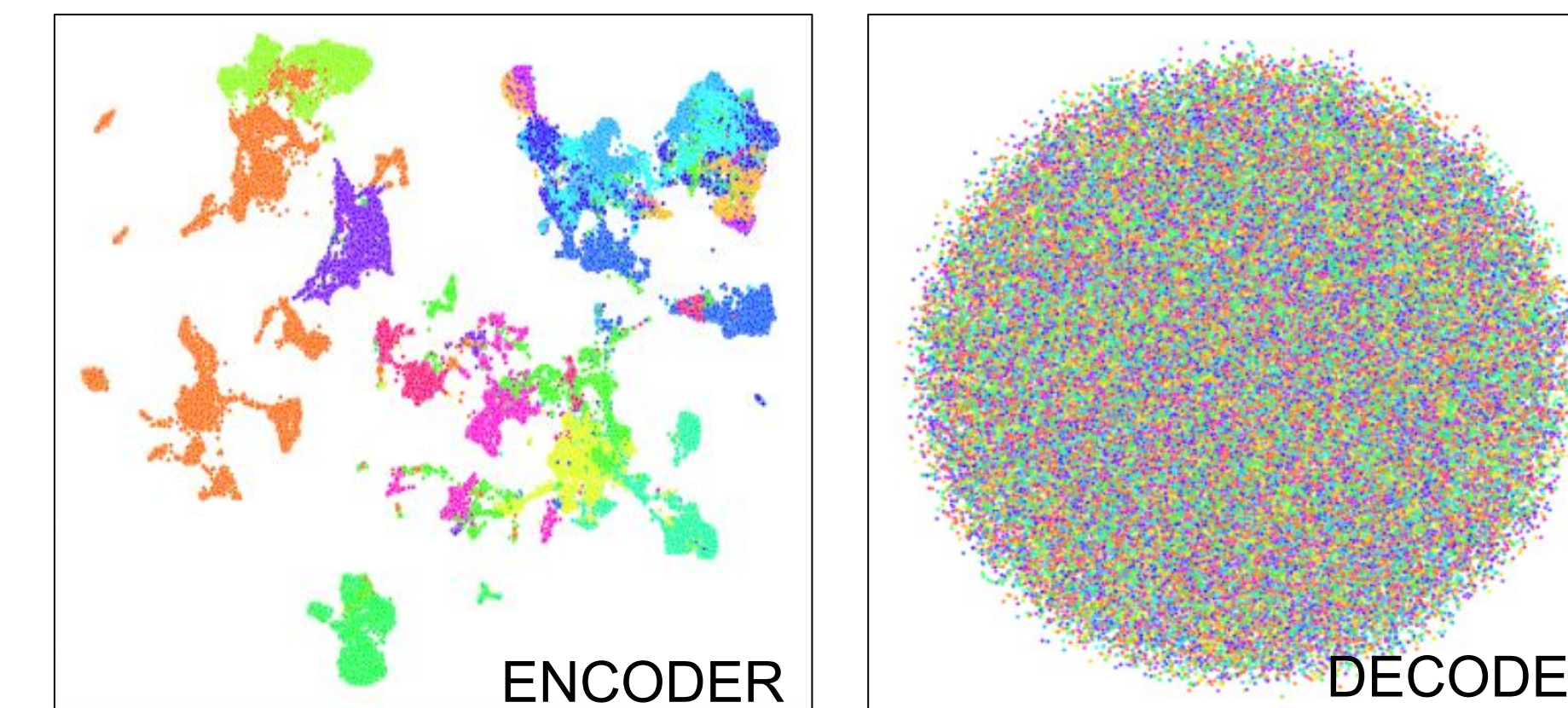
Adding depth did not appear to significantly improve loss metrics.

Step 5: Linear Comparisons



Linear Modules outperformed attention modules.

Step 4: Test Optimal Locations and Combinations



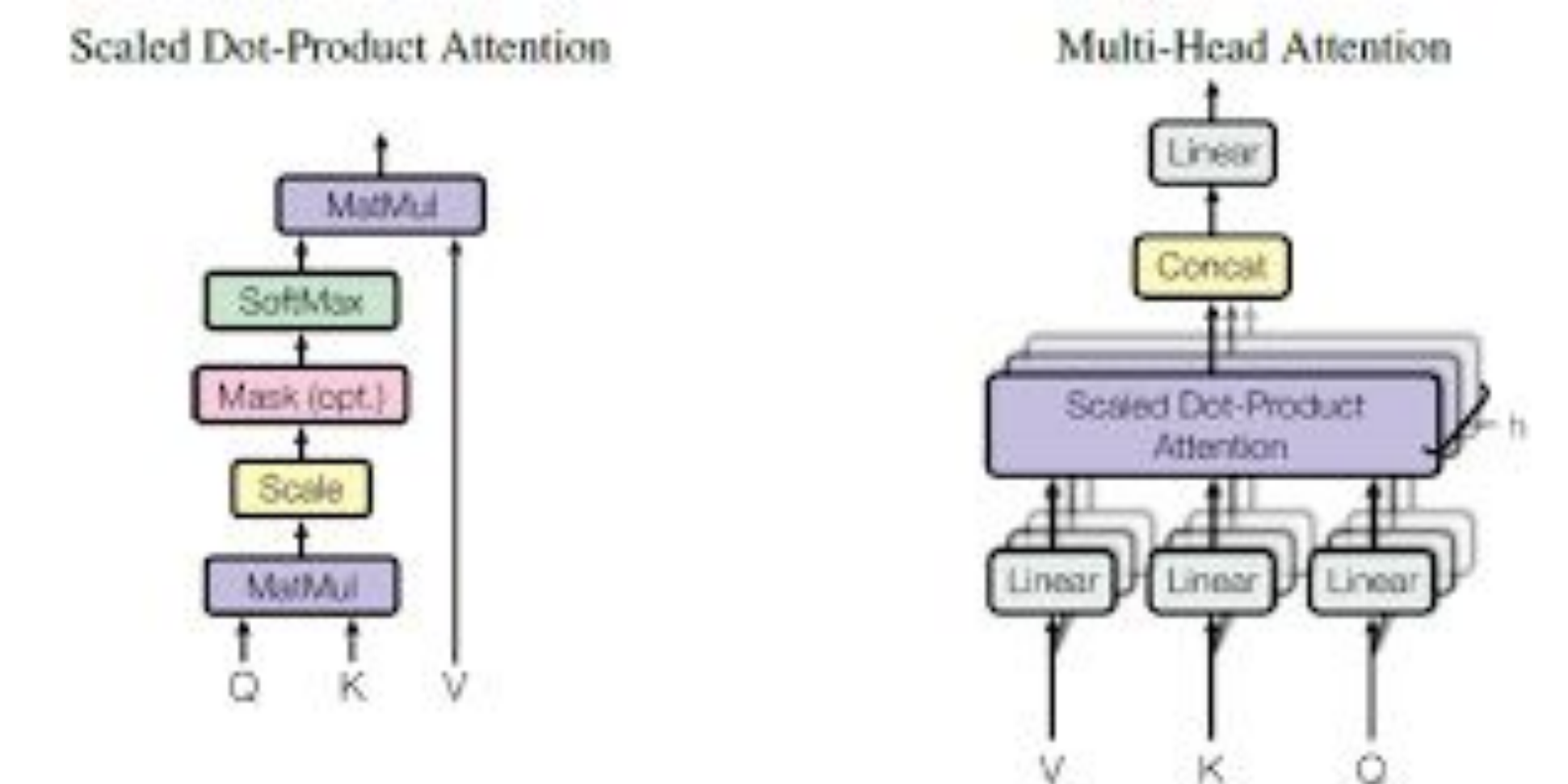
Too many modules, and decoder modules, harmed UMAPs.

Attention does not appear to outperform other methods in the compressed space of Variational Autoencoders modeling single cell counts based transcriptomics data. The linear modules improved integration and loss metrics similarly to self attention.

What is Self-Attention

	The	train	is	on	the	tracks
The	1.00	0.75	0.69	0.92	0.46	0.00
train	0.00	0.25	0.08	0.02	0.02	0.46
is	0.00	0.00	0.23	0.02	0.22	0.02
on	0.00	0.00	0.00	0.04	0.06	0.01
the	0.00	0.00	0.00	0.00	0.24	0.48
tracks	0.00	0.00	0.00	0.00	0.00	0.03

Dynamically models relationships & dependencies in sequence data.



$$sdpa(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d}})V$$

The math behind Attention

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References

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- Lopez, Romain, et al. *Deep generative modeling for single-cell transcriptomics*. Nature Methods 15.12 (2018): 1053-1058.
- Xu, Chenling, et al. *Probabilistic harmonization and annotation of single-cell transcriptomics data with deep generative models*. Molecular Systems Biology 17.1 (2021): e9620.